Q1. single-cell forward pass

Given $x_t = 0.5$, $h_{t-1} = 0.2$, $C_{t-1} = 0.4$

forget gate $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$

$f_t = \sigma(0.7(0.5) + 0.3(0.2) + 0.1)$

$f_t = \sigma(0.35 + 0.06 + 0.1) = \sigma(0.5)$

$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{0.5}} = 0.6248$

$f_t = 0.6248$

input gate $i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$

$i_t = \sigma(0.6(0.5) + 0.4(0.2) + 0)$

$i_t = \sigma(0.30 + 0.08) = \sigma(0.38)$

$i_t = 0.5939$

Candidate cell state $\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$

$\tilde{C}_t = \tanh(0.9(0.5) + 0.2(0.2) + 0)$

$\tilde{C}_t = \tanh(0.45 + 0.04) = \tanh(0.49)$

$\tilde{C}_t = 0.4542$

new cell state $C_t = f_t C_{t-1} + i_t \tilde{C}_t$

$C_t = (0.6248)(0.4) + (0.5939)(0.4542)$

$C_t = 0.2499 + 0.2697$

$C_t = 0.5196$

output gate $O_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$

$O_t = \sigma(0.5(0.5) + 0.5(0.2) + 0.1)$

$O_t = \sigma(0.25 + 0.10 + 0.1) = \sigma(0.45)$

$O_t = 0.6106$

new hidden state $h_t = O_t \tanh(C_t)$

$h_t = (0.6106)(0.4774)$

$h_t = 0.2915$

Q2. Parameter count

Given,

$D = 10$, $H = 32$, No. of gates = 4.

(a) Standard formulation: One bias vector per gate

Each gate has three sets of parameters:

input weights: $H \times D$

Recurrent weights: $H \times H$

Bias vector: H

Therefore, the total no. of parameters is:

$$P = 4(HD + H^2 + H)$$

$$P = 4(32 \times 10 + 32 \times 32 + 32)$$

$$P = 4(320 + 1024 + 32)$$

$$P = 4(1376) = 5504 \text{ trainable parameters.}$$

(b) PyTorch convention: Two bias vectors per gate

$$P = 4(HD + H^2 + 2H)$$

$$P = 4(32 \times 10 + 32 \times 32 + 2 \times 32)$$

$$P = 4(320 + 1024 + 64)$$

$$P = 4(1408) = 5632 \text{ trainable parameters.}$$

Q3. Vanishing gradient through the cell state.

Given the scaling factor $= f^K$

where f is the forget-value & K is no. of time steps.

(a) $f = 0.9$

After 10 steps ($K = 10$):

$$f^{10} = 0.9^{10}$$

$$= 0.3487$$

After 50 steps ($K = 50$):

$$f^{50} = 0.9^{50}$$

$$0.00515$$

(b) $K = 50$

Case 1: $f = 0.5$

$$f^{50} = 0.5^{50} = 8.88 \times 10^{-16}$$

Case 2: $f = 0.99$

$$f^{50} = 0.99^{50} = 0.6050$$

(c) When the forget-gate value is small, the gradient decreases exponentially over time, making it extremely difficult to learn dependencies 50 time steps back; when $f = 0.99$, about 60.5% of the gradient remains. Unlike a vanilla RNN, an LSTM has a cell state and learnable forget gate that can preserve information and allow gradients to ~~follow~~ flow across many time steps, helping reduce the vanishing-gradient problem, although it does not eliminate it completely.

Q4. Electricity load forecasting

Given:-

No. of years = 3, Hours per day = 24, Lookback window = 168 hrs, prediction horizon = 24 hrs, Features per hour = 3, Batch size = 64.

(a) No. of hourly records

Assuming 365 ~~yes~~ days per year:-

$$N = 3 \times 365 \times 24$$

$$N = 3 \times 8760$$

$$N = 26,280 \text{ records}$$

(b) No. of input/output windows:-

Each window requires 168 input hours and the following 24 output hours. With a stride of 1 hour-

$$\text{windows} = N - \text{lookback} - \text{horizon} + 1$$

$$= 26280 - 168 - 24 + 1$$

$$= 26089 \text{ windows.}$$

(c) Tensor shapes for a batch of 64 windows.

The input contains 168 time steps & 3 features per time step.

Input batch shape: (64, 168, 3)

The output contains 24 predicted hourly load values per window

Output batch shape:- (64, 24)

(d) Training windows and mini-batches

$N_{train} = 0.70 \times 26280$

$N_{train} = 18396$ records.

$W_{train} = 18396 - 168 - 24 + 1$

$W_{train} = 18205$ windows

no. of mini-batches: $B = \left\lceil \frac{18204}{64} \right\rceil$

$B = \lceil 284.453125 \rceil$

$B = 285$ mini-batches.

The final mini-batch contains 29 windows, since

$18205 - (284 \times 64) = 29$.

Assumption:- The first 70% of records from the training segment and windows are generated entirely within that segment to avoid crossing the training/validation boundary.

(e) Trainable parameters of the LSTM layer

$D = 3, H = 64$, Gates = 4, One bias per gate.

$P = 4(HD + H^2 + H)$

$P = 4(64 \times 3 + 64^2 + 64)$

$P = 4(192 + 4096 + 64)$

$P = 4(4352) = 17408$ parameters.

Q5. Predictive maintenance.

(a) Min-max normalisation:-

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$X_{norm} = \frac{95 - 20}{120 - 20} = \frac{75}{100}$$

$$X_{norm} = 0.75$$

(b) Failure prediction:-

Given:

Predicted failure probability = 0.83

Decision threshold = 0.5

Decision rule:-

if probability > 0.5 ; predict failure.

if probability < 0.5, predict no failure.

since 0.83 > 0.5, Prediction: Failure

(c) Confusion matrix & Evaluation metrics

Given:

Total cycles = 1000, Actual failures = 40

Predicted failures = 45; correctly predicted failures = 30

Step 1:- Calculate TP, TN, FP, FN

TP = 30 ; ~~FP = 40 - 30 = 1~~ FN = 40 - 30 = 10

FP = 45 - 30 = 15 ; TN = 1000 - 30 - 15 - 10 = 945

	Predicted failure	Predicted no fail
Actual failure	TP = 30	FN = 10
Actual no failure	FP = 15	TN = 945

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{30}{30+15} = \frac{30}{45} = 66.67\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{30}{30+10} = \frac{30}{40} = 75\%$$

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.6667 \times 0.75}{0.6667 + 0.75}$$

F1 score = 70.59%

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} = \frac{975}{1000} = 97.5\%$$

(d) The maintenance ~~team~~ team should prioritise recall, because missing an actual pump failure (a false negative) can cause costly equipment damage, unexpected downtime, or safety risks. Accuracy is potentially misleading because failures are rare (only 40 out of 1000 cycles), so a model could achieve high accuracy by predicting no failure for almost every cycle while missing many real failures.